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Studierichting: Informatica
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Opgave: Bepaal de kritieke punten en ga na of de functie in die punten een extremum bereiken:

$$f(x, y) = 2y^2 - x(x - 1)^2$$

$$\frac{\partial f}{\partial x} = \frac{d}{dx} (-x(x - 1)^2) = \frac{d}{dx} (-x^3 + 2x^2 - x) = -3x^2 + 4x - 1$$

$$\frac{\partial f}{\partial y} = \frac{d}{dy} (2y^2) = 4y$$

$$\Rightarrow \begin{cases} -3x^2 + 4x - 1 & = & 0 \\ 4y & = & 0 \end{cases} \Rightarrow \begin{cases} 3x^2 - 4x + 1 & = & 0 \\ y & = & 0 \end{cases}$$

$$D = 16 \cdot -4 \cdot 3 = 4 \Rightarrow x_{1,2} = \frac{4 \pm 2}{6} \Rightarrow x_1 = \frac{1}{3} \text{ en } x_2 = 1$$

$$\Rightarrow \text{Kritieke punten: } \left(\frac{1}{3}, 0\right) \text{ en } (1, 0)$$

$$H(x, y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} = \begin{pmatrix} -6x + 4 & 0 \\ 0 & 4 \end{pmatrix}$$

$$\Delta(x, y) = \det H(x, y) = (-6x + 4) \cdot 4 = -24x + 16$$

$$\Delta\left(\frac{1}{3}, 0\right) = -\frac{24}{3} + 16 = -8 + 16 = 8 > 0 \Rightarrow \text{minimum of maximum}$$

$$\Rightarrow A\left(\frac{1}{3}, 0\right) = -\frac{6}{3} + 4 = -2 + 4 = 2 > 0 \Rightarrow \text{minimum}$$

$$\Delta(1, 0) = -24 + 16 = -8 < 0 \Rightarrow \text{zadelpunt}$$

References